Module 03

1. Define network layer main function performed by network layer. (5 marks)

The main functions performed by the network layer are:
Routing: When a packet reaches the router's input link, the router will move the packets to the router's output link. For example, a packet from S1 to R1 must be forwarded to the next router on the path to S2.
Logical Addressing: The data link layer implements the physical addressing and network layer implements the logical addressing. Logical addressing is also used to distinguish between source and destination system. The network layer adds a header to the packet which includes the logical addresses of both the sender and the receiver.
Internetworking: This is the main role of the network layer that it provides the logical connection between different types of networks. Fragmentation: The fragmentation is a process of breaking the packets into the smallest individual data units that travel through different network

2. Difference between routing and forwarding. (5 marks)

Routing is the process of moving data from one device to another device. These are two other services offered by the network layer. In a network, there are a number of routes available from the source to the destination. The network layer specifies some strategies which find out the best possible route. This process is referred to as routing. There are a number of routing protocols that are used in this process and they should be run to help the routers coordinate with each other and help in establishing communication throughout the net

Forwarding is simply defined as the action applied by each router when a packet arrives at one of its interfaces. When a router receives a packet from one of its attached networks, it needs to forward the packet to another attached network (unicast routing) or to some attached networks (in the case of multicast routing). Routers are used on the network for forwarding a packet from the local network to the remote network. So, the process of routing involves packet forwarding from an entry interface out to an exit inter

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3. Virtual Circuit in Networking

A **virtual circuit** is a logical, predefined path established for communication in a packet-switched network. Unlike traditional circuits, which require dedicated physical resources, virtual circuits use the same network resources but simulate a dedicated connection between two endpoints. The virtual circuit ensures that all packets travel along the same predetermined route in the correct order, making it reliable and efficient for data transmission. It's commonly used in protocols such as X.25, Frame Relay, and ATM (Asynchronous Transfer Mode).

Types of Virtual Circuits

1. **Switched Virtual Circuit (SVC)**:
   - **Definition**: Dynamically established for temporary use.
   - **Setup**: The circuit is set up only when required and terminated after the data transfer is complete.
   - **Use Cases**: Suitable for intermittent communication needs.
2. **Permanent Virtual Circuit (PVC)**:
   - **Definition**: Predefined and always available for communication.
   - **Setup**: The circuit is established once and maintained indefinitely.
   - **Use Cases**: Suitable for continuous communication between two endpoints.

Key Characteristics of Virtual Circuits:

- **Connection-Oriented**: A logical path is established before data transmission begins.
- **Reliable**: Ensures packets arrive in the correct order and with minimal loss.
- **Sequence Numbers**: Used to maintain the correct order of packets.
- **Fixed Path**: Packets travel through the same route, optimizing efficiency and reducing routing decisions.

Advantages of Virtual Circuits:

1. **Efficient Resource Use**: Resources are reserved only when needed.
2. **Reliable Delivery**: Guarantees in-sequence delivery of packets.
3. **Optimized Routing**: Fixed paths reduce congestion and latency.
4. **Predictable Performance**: SVCs can offer better performance for bursty traffic, while PVCs offer stable, continuous performance.

Disadvantages of Virtual Circuits:

1. **Overhead**: The setup phase can introduce delays.
2. **Resource Reservation**: PVCs may reserve resources unnecessarily, leading to inefficiencies.
3. **Limited Scalability**: Scaling SVCs might require frequent setup/teardown phases.
4. **Failure Impact**: Any failure in the path can disrupt the entire communication session.

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4. Virtual Circuit Model Components with Commands

The **Virtual Circuit model** works in three distinct phases: Setup, Data Transfer, and Teardown.

1. Setup Phase:

- **Objective**: Establishes a logical path between the sender and receiver.
- **Actions**:
  - The sender contacts the network layer and specifies the receiver's address.
  - The network layer computes the best path (sequence of links and switches).
  - Routing tables in intermediate nodes are updated to reflect this path.
  - The network may reserve bandwidth along this path.
- **Example Command**: `SETUP VIRTUAL CIRCUIT <Sender Address> <Receiver Address>`.

2. Data Transfer Phase:

- **Objective**: Data packets are transmitted along the established path.
- **Actions**:
  - Data packets carry a Virtual Circuit Identifier (VCI) rather than a full address.
  - The VCI helps routers and switches identify the correct path and packet sequence.
- **Example Command**: `SEND DATA <VCI> <Data>`.

3. Teardown Phase:

- **Objective**: Terminates the logical connection and releases reserved resources.
- **Actions**:
  - The sender or receiver requests the termination.
  - Routing tables in the network are updated, and resources are freed.
  - Signaling messages indicate the termination to both endpoints and intermediate nodes.
- **Example Command**: `TEARDOWN VIRTUAL CIRCUIT <VCI>`.

5. Working Procedure of Datagram Services Model in the Network (10 Marks)

The **Datagram Service Model** is a connectionless, best-effort method of communication where data is sent in discrete units (datagrams). Unlike virtual circuit-based communication, datagrams are transmitted without establishing a dedicated path beforehand. Each datagram is routed independently through the network, potentially taking different routes to the destination.

Detailed Working Procedure:

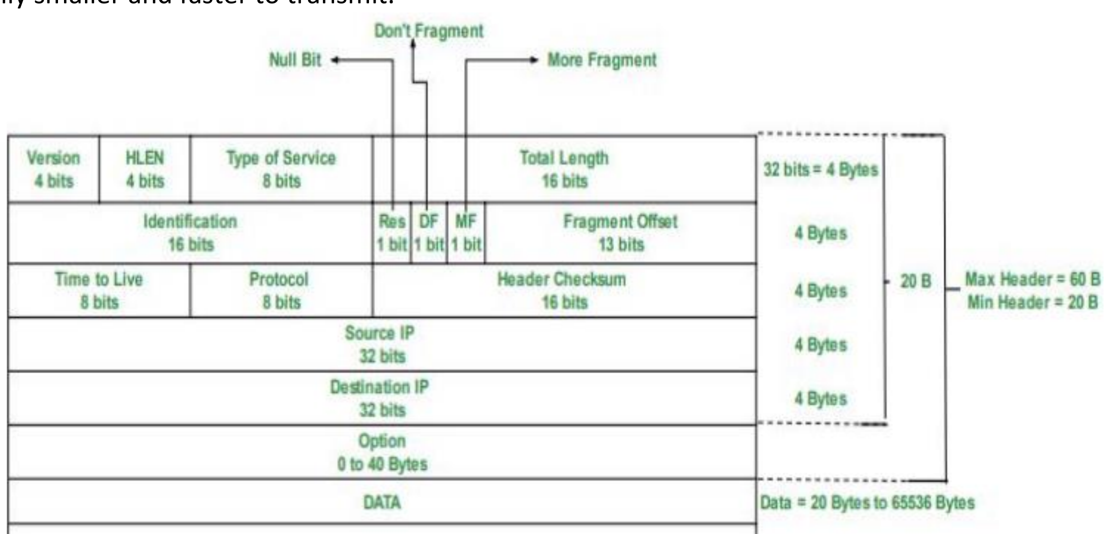
1. **Packetization**:
    - The sending device divides data into smaller units called datagrams. Each datagram contains its own header and payload. The payload includes the actual data being transmitted, while the header contains control information (e.g., source and destination IP addresses, packet length).
2. **Addressing**:
    - Each datagram carries addressing information in its header, which includes the source and destination IP addresses. This enables the network devices (routers and switches) to know where the datagram is coming from and where it's going.
3. **Routing**:
    - Each router along the path makes an independent decision about how to forward the datagram based on its current routing table. These routing decisions are made dynamically and may vary from one datagram to another. Routers use algorithms such as distance-vector or link-state to determine the best path for forwarding.
4. **No Guarantee of Delivery**:
    - In the datagram model, there is no guarantee that a datagram will arrive at the destination. Datagrams may be lost due to congestion, network failures, or other issues. Moreover, if a datagram is delayed, it might not arrive in the order it was sent.
5. **Destination Handling**:
    - Upon receiving the datagram, the destination device uses the addressing information to deliver the payload to the appropriate application or service. If any packets are missing or out of order, the receiver may request retransmission or rely on higher-layer protocols like TCP to handle these issues.

Examples of Datagram Service Models:

- **IPv4** and **IPv6**: The most common datagram-based protocols.
- **UDP (User Datagram Protocol)**: A connectionless transport-layer protocol, uses datagrams to send data.

Key Characteristics:

- **Connectionless**: No prior connection setup.
- **Independent Routing**: Each datagram is routed independently.
- **Best-Effort**: There is no guarantee of delivery or sequence.
- **Low Overhead**: Due to the absence of connection setup and maintenance, datagrams are typically smaller and faster to transmit.



6. Difference Between Virtual Circuit and Datagram Network (10 Marks)

Aspect	Virtual Circuit Network	Datagram Network
Connection Type	Connection-oriented, requires a logical path setup before transmission	Connectionless, no need to set up a path before transmission
Path Setup	Path is established beforehand (either permanent or switched)	No predefined path; each packet is routed independently
Routing	Fixed path determined at setup; all packets follow the same route	Independent routing for each packet; may take different paths
Reliability	Reliable, guarantees delivery in order (especially for PVC)	No guarantee of delivery or order of packets
Flow Control	Managed by the network (flow control is inherent in the setup)	Not inherent; flow control is the responsibility of higher layers (e.g., TCP)
Congestion Handling	Some protocols like ATM or Frame Relay have built-in congestion control	No inherent congestion control; relies on higher-layer protocols
Setup Time	Requires time for setup (more overhead)	Immediate; no setup time required
Example Protocols	X.25, ATM, Frame Relay	IP, UDP, raw Internet communication
Error Detection	Typically provides error detection and recovery at the network layer	Error handling is handled by higher layers, like TCP
Network Overhead	Higher overhead due to path establishment and resource allocation	Lower overhead due to lack of connection management

Connectionless vs. Connection-Oriented Communication:

- **Connection-Oriented (Virtual Circuit)**: Involves a formal connection establishment phase, packet ordering, and teardown. It ensures data reliability and correct sequence.
- **Connectionless (Datagram)**: Involves no setup, each packet is treated independently, and there's no guarantee of reliability or ordering.

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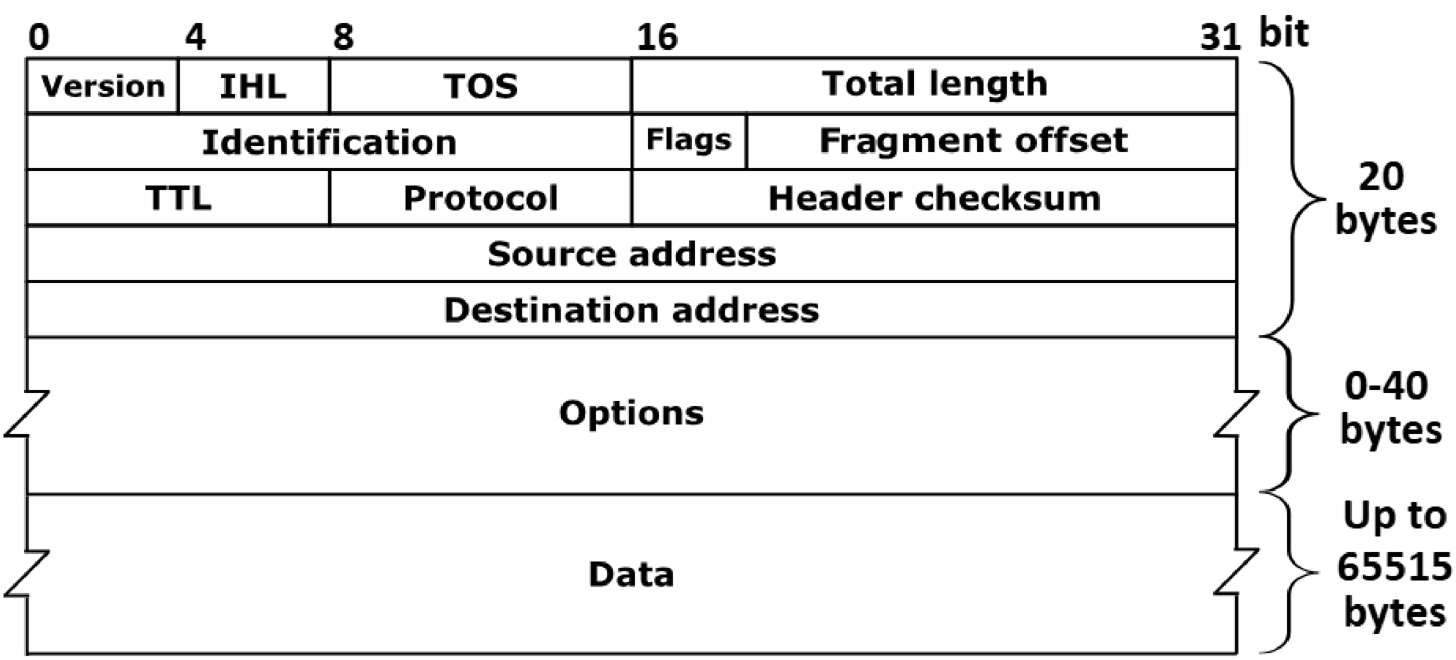
7. Diagram of IPv4 Datagram Format with Brief Explanation (10 Marks)



Explanation:

- **Version (4 bits)**: Specifies the version of IP (IPv4 = 4).
- **IHL (Internet Header Length, 4 bits)**: Specifies the length of the header in 32-bit words.
- **Type of Service (8 bits)**: Defines the quality of service for the packet (used for prioritization).
- **Total Length (16 bits)**: Specifies the total length of the datagram (header + data).
- **Identification (16 bits)**: Used for fragmenting large packets, ensuring they can be reassembled at the destination.
- **Flags (3 bits) and Fragment Offset (13 bits)**: Used for fragmentation and reassembly.
- **TTL (8 bits)**: Time to live, used to prevent packets from circulating forever.
- **Protocol (8 bits)**: Identifies the next-level protocol (e.g., TCP, UDP).
- **Header Checksum (16 bits)**: Ensures header integrity by detecting errors in the header.
- **Source and Destination Address (32 bits each)**: IP addresses of the sender and receiver.
- **Options**: Optional field for additional routing or security information.
- **Data**: The actual payload or user data.

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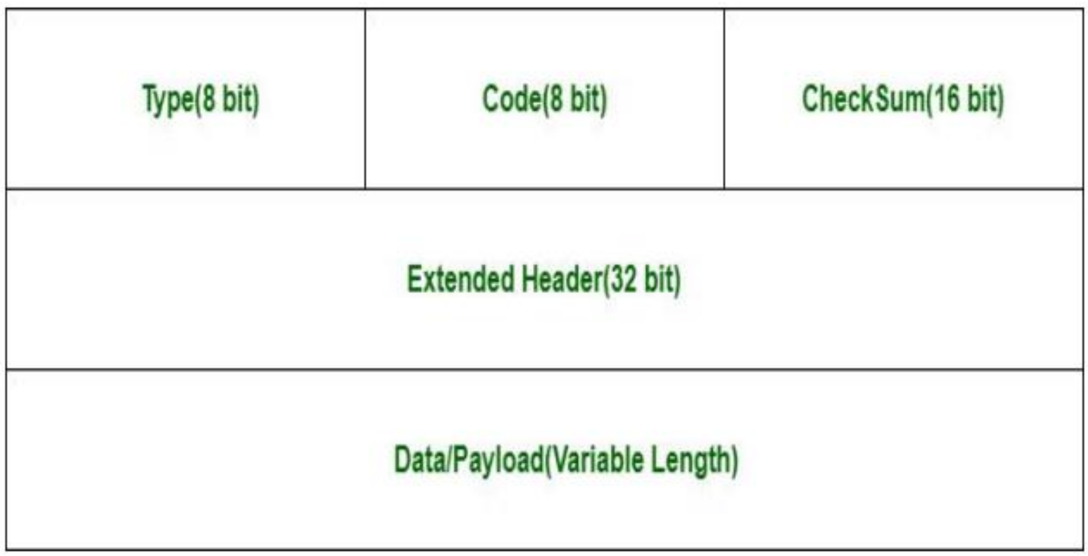
8. Diagram and Explanation of ICMP (10 Marks)

ICMP Packet Format

ICMP header comes after IPv4 and IPv6 packet header.



Explanation:

- **Type (8 bits)**: Specifies the type of ICMP message (e.g., Echo Request, Echo Reply).
- **Code (8 bits)**: Provides further context for the message type (e.g., Error Message).
- **Checksum (16 bits)**: Used for error detection in the ICMP message.

Uses of ICMP:

1. **Error Reporting**: ICMP messages inform hosts about network errors like unreachable destinations or time exceeded.
2. **Diagnostics**: Tools like `ping` (Echo Request) and `traceroute` use ICMP to diagnose network issues.
3. **Path MTU Discovery**: ICMP helps in determining the maximum packet size that can be transmitted without fragmentation.
4. **Network Feedback**: Provides feedback about the state of the network or the path.

Working Procedure:

- The sender generates an ICMP Echo Request message.
- The network delivers the message to the destination.
- The destination replies with an Echo Reply message if it is reachable.
- The message is transmitted back to the sender for diagnostics.

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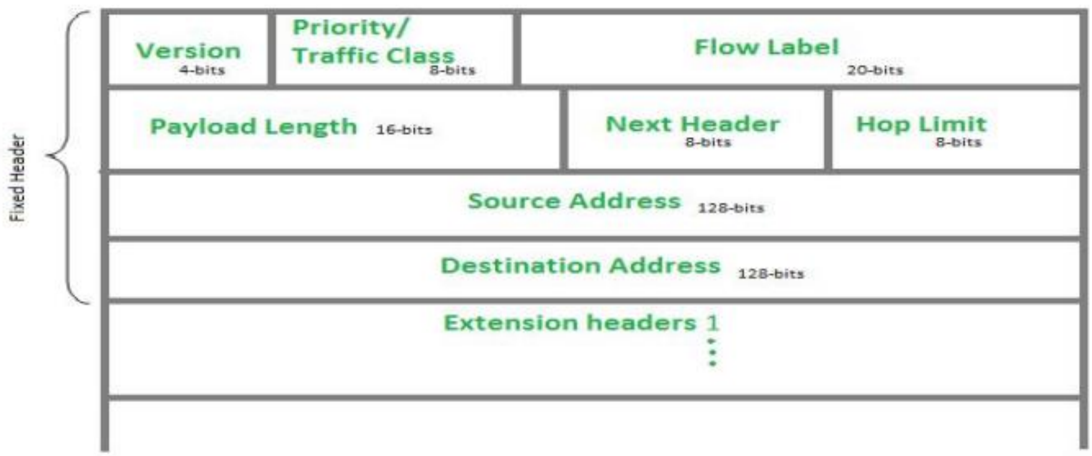
9. Diagram of IPv6 Header Format with Brief Explanation (10 Marks)



Explanation:

- **Version (4 bits)**: IPv6 version (6).
- **Traffic Class (8 bits)**: Differentiates traffic for QoS (Quality of Service).
- **Flow Label (20 bits)**: Identifies packets belonging to the same flow, used for efficient processing.
- **Payload Length (16 bits)**: Length of the data portion.
- **Next Header (8 bits)**: Indicates the protocol type for the next header (e.g., TCP, UDP).
- **Hop Limit (8 bits)**: Similar to TTL in IPv4, it limits the number of hops before the packet is discarded.
- **Source and Destination Address (128 bits each)**: The sender's and receiver's IPv6 addresses.
- **Data**: The payload, or actual data being transferred.

10. What's Inside a Router? Components for Router Identification (10 Marks)

A **router** is a networking device responsible for forwarding data packets between networks. It examines packets' IP addresses to determine the best path for forwarding.

Components of a Router:

1. **Central Processor (CPU)**: Executes routing protocols, handles packet forwarding, and controls the router's functions.
2. **RAM**: Temporarily stores routing tables, forwarding tables, and incoming packet data.
3. **ROM/Flash Memory**: Stores firmware and operating systems that control the router.
4. **Network Interfaces**: Physical or virtual ports for connecting to different networks (Ethernet, Wi-Fi, etc.).
5. **Routing Table**: Stores the routes that the router uses to forward packets. This table is dynamically updated based on the routing protocol.
6. **Power Supply**: Provides energy to the router’s components.

Router Identification:

- **IP Address**: Each router interface has an IP address for communication with other devices.
- **MAC Address**: The router's physical hardware address used for Layer 2 communication.
- **Routing Protocols**: Protocols like OSPF, BGP, or RIP determine the best path and help identify the router's role in the network.

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11. Congestion Control Techniques in Network Layer (10 Marks)

Congestion Control refers to techniques designed to prevent or alleviate congestion in a network, which can lead to packet loss, delays, and degraded performance.

Techniques:

1. **Flow Control**:
    - Manages the rate at which data is sent between devices to avoid overloading the network. It ensures the sender does not overwhelm the receiver or intermediate devices.
2. **Traffic Shaping**:
    - Adjusts the traffic entering the network by limiting the rate at which data is sent. Traffic shaping can smooth traffic bursts, making the network more predictable.
3. **Random Early Detection (RED)**:
    - Used by routers to preemptively drop packets before queues overflow, signaling the sender to slow down. RED helps avoid global congestion by reducing the packet loss rate.
4. **TCP Congestion Control**:
    - **Slow Start**: Initially increases the transmission rate slowly, and then accelerates when the network's capacity is detected.
    - **Congestion Avoidance**: Gradually reduces the transmission rate when network congestion is suspected.
    - **Fast Retransmit and Fast Recovery**: Quickly retransmits lost packets to avoid further congestion.
5. **Queue Management**:
    - Routers manage queues to control congestion. Techniques like **First-In, First-Out (FIFO)** or **Priority Queuing** help to prioritize important traffic while preventing network bottlenecks.